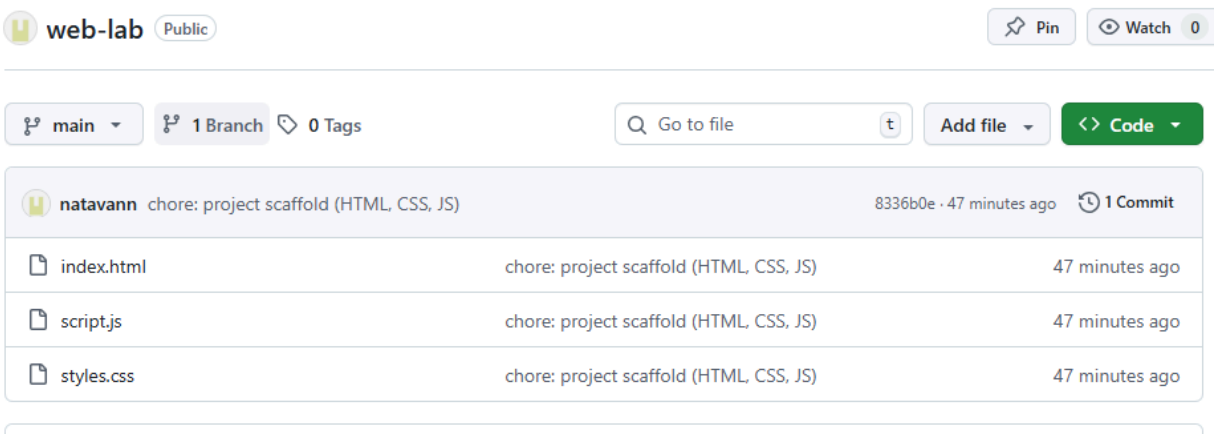
Lab 2

```
natavan@DESKTOP-GSHQS04:~$ mkdir web-lab && cd web-lab
natavan@DESKTOP-GSHQS04:~/web-lab$ code .
natavan@DESKTOP-GSHQS04:~/web-lab$ touch index.html styles.css script.jss
natavan@DESKTOP-GSHQS04:~/web-lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/natavan/web-lab/.git/
natavan@DESKTOP-GSHQS04:~/web-lab$ git add .
natavan@DESKTOP-GSHQS04:~/web-lab$ git commit -m "chore: project scaffold (HTML, CSS, JS)"
[master (root-commit) 8336b0e] chore: project scaffold (HTML, CSS, JS)
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 index.html
 create mode 100644 script.js
 create mode 100644 styles.css
natavan@DESKTOP-GSHQS04:~/web-lab$ git re
rebase         reflog         remote         repack         replace        request-pull   reset          restore        revert
natavan@DESKTOP-GSHQS04:~/web-lab$ git remote add origin https://github.com/natavann/DV-FT-MAR2026.git
natavan@DESKTOP-GSHQS04:~/web-lab$ git branch -m main
```

```
natavan@DESKTOP-GSHQS04:~/web-lab$ git push --set-upstream origin main
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
To https://github.com/natavann/DV-FT-MAR2026.git
 ! [rejected]        main -> main (non-fast-forward)
error: failed to push some refs to 'https://github.com/natavann/DV-FT-MAR2026.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
natavan@DESKTOP-GSHQS04:~/web-lab$ git remote -v
origin  https://github.com/natavann/DV-FT-MAR2026.git (fetch)
origin  https://github.com/natavann/DV-FT-MAR2026.git (push)
natavan@DESKTOP-GSHQS04:~/web-lab$ git remote set-url origin https://github.com/natavann/web-lab.git
natavan@DESKTOP-GSHQS04:~/web-lab$ git push --set-upstream origin main
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 263 bytes | 10.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/natavann/web-lab.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
natavan@DESKTOP-GSHQS04:~/web-lab$
```



Exercise 1

```
natavan@DESKTOP-GSHQS04:~/web-lab$ git push -u origin develop
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/natavann/web-lab/pull/new/develop
remote:
To https://github.com/natavann/web-lab.git
 * [new branch]      develop -> develop
branch 'develop' set up to track 'origin/develop'.
natavan@DESKTOP-GSHQS04:~/web-lab$ git checkout -b feature/header-banner
Switched to a new branch 'feature/header-banner'
natavan@DESKTOP-GSHQS04:~/web-lab$ git add .
natavan@DESKTOP-GSHQS04:~/web-lab$ git commit -m "feat: add header banner with nav links"
On branch feature/header-banner
nothing to commit, working tree clean
natavan@DESKTOP-GSHQS04:~/web-lab$ git push -u origin feature/header-banner
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature/header-banner' on GitHub by visiting:
remote:      https://github.com/natavann/web-lab/pull/new/feature/header-banner
remote:
To https://github.com/natavann/web-lab.git
 * [new branch]      feature/header-banner -> feature/header-banner
branch 'feature/header-banner' set up to track 'origin/feature/header-banner'.
natavan@DESKTOP-GSHQS04:~/web-lab$
```

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0"
/>
    <title>Git Flow Lab</title>
```

```html
    <link rel="stylesheet" href="styles.css" />
  </head>
  <body>
    <header class="navbar">
      <nav>
        <div class="logo">DevOps Lab</div>
        <ul>
          <li><a href="#">Home</a></li>
          <li><a href="#">Projects</a></li>
          <li><a href="#">Contact</a></li>
        </ul>
      </nav>
    </header>

    <main>
      <h1>Hello, Git Flow 👋</h1>
    </main>

    <script src="script.js"></script>
  </body>
</html>
```

```css
body {
  font-family: system-ui, sans-serif;
  margin: 0;
  padding: 0; /* Changed from 2rem to 0 so navbar touches the edges */
  background: #f6f8fa;
}

/* Navbar Styling */
.navbar {
  background-color: #24292f; /* Dark GitHub-style grey */
  color: white;
  padding: 1rem 2rem;
}

.navbar nav {
  display: flex;
  justify-content: space-between;
  align-items: center;
```

```
}

.navbar ul {
  list-style: none;
  display: flex;
  margin: 0;
  padding: 0;
  gap: 20px; /* Space between links */
}

.navbar a {
  color: white;
  text-decoration: none;
  font-weight: 500;
}

.navbar a:hover {
  text-decoration: underline;
}

/* Keep your main content centered */
main {
  padding: 2rem;
}
```

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
main	47 minutes ago		Default		

Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/header-banner	2 minutes ago		0	0	
develop	36 minutes ago		0	0	

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/header-banner	2 minutes ago		0	0	
develop	36 minutes ago		0	0	

```
natavan@DESKTOP-GSHQS04:~/web-lab$ git remote -v
origin  https://github.com/natavann/web-lab.git (fetch)
origin  https://github.com/natavann/web-lab.git (push)
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch feature/header-banner
Your branch is up to date with 'origin/feature/header-banner'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html
        modified:   styles.css

no changes added to commit (use "git add" and/or "git commit -a")
natavan@DESKTOP-GSHQS04:~/web-lab$ git add .
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch feature/header-banner
Your branch is up to date with 'origin/feature/header-banner'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   index.html
        modified:   styles.css

natavan@DESKTOP-GSHQS04:~/web-lab$ git commit -m "feat: add header banner with nav links"
[feature/header-banner 53b9f59] feat: add header banner with nav links
 2 files changed, 69 insertions(+)
natavan@DESKTOP-GSHQS04:~/web-lab$ git push -u origin feature/header-banner
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1019 bytes | 29.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/natavann/web-lab.git
   8336b0e..53b9f59  feature/header-banner -> feature/header-banner
branch 'feature/header-banner' set up to track 'origin/feature/header-banner'.
natavan@DESKTOP-GSHQS04:~/web-lab$
```

Exercise 2

```
natavan@DESKTOP-GSHQS04:~/web-lab$ git checkout -b feature/cta-button
Switched to a new branch 'feature/cta-button'
natavan@DESKTOP-GSHQS04:~/web-lab$ git add .
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch feature/cta-button
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   index.html
        modified:   script.js
        modified:   styles.css

natavan@DESKTOP-GSHQS04:~/web-lab$ vim README.md
natavan@DESKTOP-GSHQS04:~/web-lab$ git add .
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch feature/cta-button
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   README.md
        modified:   index.html
        modified:   script.js
        modified:   styles.css
```

Style.css file

```
/* --- New Styles for Feature: CTA Button --- */

.cta-btn {
  background-color: #0969da;
  color: white;
  border: none;
  padding: 8px 16px;
  border-radius: 6px;
  cursor: pointer;
  font-weight: 600;
  transition: background 0.2s;
}

.cta-btn:hover {
  background-color: #054da5;
}

/* --- Dark Theme logic --- */

/* When data-theme is dark, change the body background and text */
[data-theme="dark"] {
  background-color: #0d1117; /* GitHub dark mode color */
  color: #f0f6fc;
}

/* Make the navbar slightly different in dark mode */
[data-theme="dark"] .navbar {
  background-color: #161b22;
  border-bottom: 1px solid #30363d;
}

/* Ensure links stay white or visible */
[data-theme="dark"] .navbar a {
  color: #58a6ff;
}
```

```javascript
// script.js
console.log("JS is connected! 🚀");

const themeToggle = document.getElementById('theme-toggle');

themeToggle.addEventListener('click', () => {
  // Check if body currently has the 'dark' theme
  const isDark = document.body.getAttribute('data-theme') === 'dark';

  // Flip the attribute
  document.body.setAttribute('data-theme', isDark ? 'light' : 'dark');

  // Update button text so the user knows what happened
  themeToggle.textContent = isDark ? '🌙 Dark Mode' : '☀️ Light Mode';
});
```

index.html

```html
<li><a href="#">Home</a></li>
          <li><a href="#">Projects</a></li>
          <li><a href="#">Contact</a></li>
          <li><button id="theme-toggle" class="cta-btn">🌙 Dark
Mode</button></li>
```

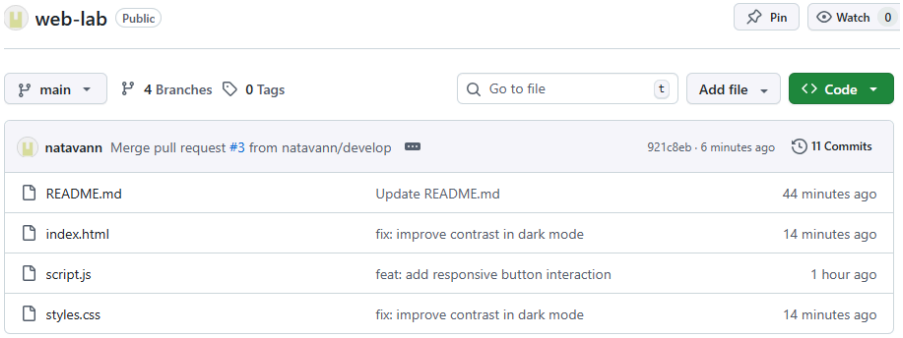
Exercise 3

```
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
natavan@DESKTOP-GSHQS04:~/web-lab$ git pull origin main
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.86 KiB | 190.00 KiB/s, done.
From https://github.com/natavann/web-lab
 * branch            main       -> FETCH_HEAD
   8336b0e..d6ccaa4  main       -> origin/main
Updating 8336b0e..d6ccaa4
Fast-forward
 README.md   |  5 +++++
 index.html  | 31 ++++++++++++++++++++++++++++++
 script.js   | 15 +++++++++++++++
 styles.css  | 78 ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 4 files changed, 129 insertions(+)
 create mode 100644 README.md
natavan@DESKTOP-GSHQS04:~/web-lab$ git checkout -b fix/contrast-text
Switched to a new branch 'fix/contrast-text'
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch fix/contrast-text
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html
        modified:   styles.css

no changes added to commit (use "git add" and/or "git commit -a")
natavan@DESKTOP-GSHQS04:~/web-lab$ git add .
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch fix/contrast-text
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   index.html
        modified:   styles.css
```

Edited index.html file:

```
<li><button id="theme-toggle" class="cta-btn">Toggle High-Contrast
Theme</button></li>
```

Edited styles.css file:

```
/* Update this specific block in your CSS */
[data-theme="dark"] {
  background-color: #0d1117;
  color: #ffffff; /* Pure white for maximum contrast */
}

[data-theme="dark"] .navbar a {
  color: #79c0ff; /* Brighter blue for better visibility */
  text-shadow: 0 0 2px rgba(0,0,0,0.5); /* Adds a "pop" for readability */
```

```
natavan@DESKTOP-GSHQS04:~/web-lab$ git fetch origin
natavan@DESKTOP-GSHQS04:~/web-lab$ git pull --rebase origin develop
From https://github.com/natavann/web-lab
 * branch            develop    -> FETCH_HEAD
Successfully rebased and updated refs/heads/develop.
natavan@DESKTOP-GSHQS04:~/web-lab$ git push origin develop
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 4 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (13/13), 2.44 KiB | 277.00 KiB/s, done.
Total 13 (delta 5), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (5/5), completed with 1 local object.
To https://github.com/natavann/web-lab.git
   90775a9..76f8b52  develop -> develop
natavan@DESKTOP-GSHQS04:~/web-lab$ git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.
natavan@DESKTOP-GSHQS04:~/web-lab$ git status
On branch develop
Your branch is up to date with 'origin/develop'.

nothing to commit, working tree clean
natavan@DESKTOP-GSHQS04:~/web-lab$ git merge main
Merge made by the 'ort' strategy.
natavan@DESKTOP-GSHQS04:~/web-lab$ git push
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.13 KiB | 1.07 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/natavann/web-lab.git
   76f8b52..a8a2325  develop -> develop
```